Mainframe Training VM

Setup Guide: AWS EC2 (m7i-flex.large, Windows) + Remote Desktop + Developer Tools

By the end of this guide you will have a Windows cloud workstation that you reach with Remote Desktop and that has **Git, GnuCOBOL, Python 3, Node.js, Claude Code and VS Code** installed, ready for the COBOL and Claude Code training labs. Each step says what to do, why you are doing it, and what you should see on screen.

What you will do

Part	What happens	Result
1	Create an AWS account	A secured account with a spending alert
2	Launch an EC2 instance	A running Windows VM (m7i-flex.large)
3	Connect with Remote Desktop	You are logged in to the VM desktop
4	Install the six tools	Git, GnuCOBOL, Python 3, Node.js, Claude Code, VS Code
5	Verify every tool	Six version numbers on screen
6	Daily care and cleanup	Stop the VM so it does not keep billing
7	Bonus (optional): sign in to Claude Code	Claude Code open and logged in on the VM

Before you start

- A **work email** (your 77GSI address) to register the AWS account. A fallback is explained in Step 1.2.
- A **mobile phone** that can receive a verification code, and a **payment method**, which AWS may ask for during sign-up.
- A **Windows PC** with Remote Desktop Connection (built in). On a Mac, install the Microsoft Remote Desktop app.
- A **password manager** (recommended) for the root password and the VM password.

Item	Value used in this guide
Cloud / service	AWS Free Plan account / EC2
Instance type	m7i-flex.large (2 vCPU, 8 GiB memory)
Operating system	Windows Server 2025 Base (2022 also works)
Access	Remote Desktop (RDP, TCP port 3389)
Tools	Git, GnuCOBOL, Python 3, Node.js, Claude Code, VS Code

Watch the cost. A Windows instance bills per hour for compute plus the Windows licence. Free Plan credits only cover this until they run out. **Stop** the instance at the end of each working day, set a budget alert (Step 1.6), and terminate the instance when the training ends.

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About the screenshots. Parts 2 and 3 use real screenshots of the AWS console and the Remote Desktop session. Parts 4, 5 and 7 use screenshots from the training VM. Your account ID, public IP and public DNS are blacked out; yours will show your own values. Never put card numbers, passwords, access keys or the contents of your .pem file in a screenshot.

1. Create the AWS account

An AWS account is the container for everything you create, including your VM. You only need to do this once.

Step 1.1 Start the sign-up

Open **https://aws.amazon.com/free** in your browser and click **Create a Free Account**. AWS then walks you through a short sign-up form. The steps below explain what each part asks for.

Step 1.2 Enter your email and verify it

Type your **root user email** and an **account name**, then click **Verify email address**. AWS emails you a code; type it in to continue. The root user is the all-powerful owner of the account, so use an email you control long term.

Which email should you use? Use your 77GSI company email, for example `jirinco@77global.biz`. One email can only be linked to one AWS account. If your 77GSI email was already used for AWS, register with a Gmail address in the form `<first-initial><lastname>77@gmail.com`, for example `jirinco77@gmail.com`.

Step 1.3 Set a strong root password

Choose a long, unique password and save it in your password manager. You will use it to sign in as the root user.

Step 1.4 Choose a plan and enter your details

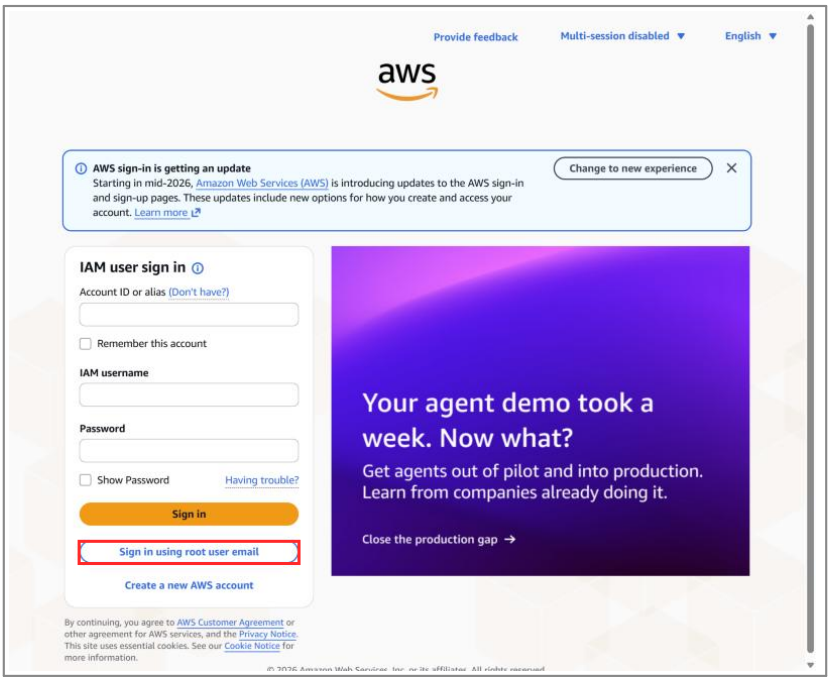
Select **Personal** or **Business** and fill in your contact details. AWS may also ask for a payment method and a phone verification (an SMS or voice code). Complete these on your own account when prompted.

Registration screens are not shown in this guide. The sign-up form, phone verification and billing pages contain personal and payment details, so no screenshots are included. Never share card numbers or codes with anyone.

Step 1.5 Pick the free plan and sign in

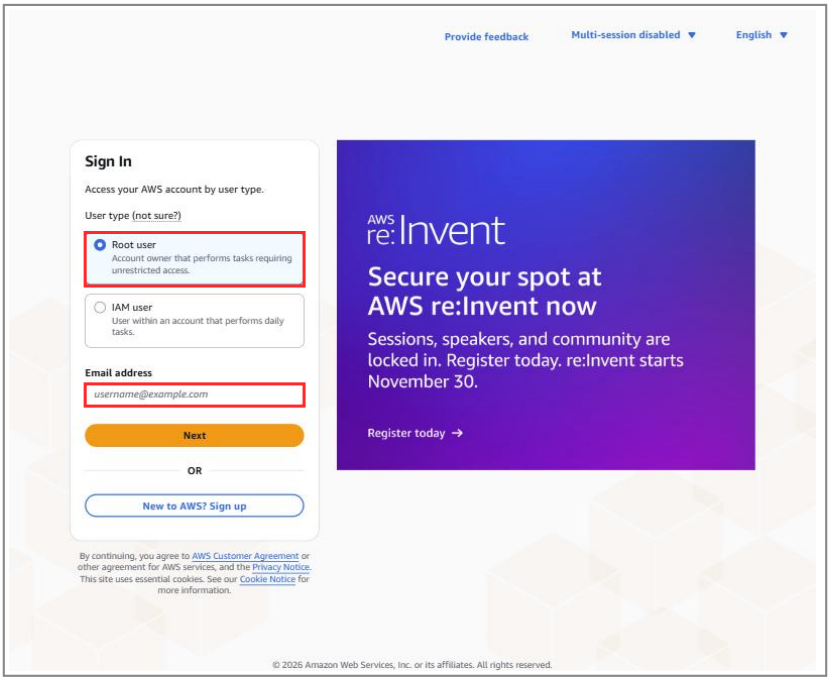
Choose the **Free** support plan and select the **Free account plan**. Activation usually takes a few minutes but can take up to 24 hours; AWS emails you when it is ready.

To sign in: open **https://console.aws.amazon.com**. The page opens on the **IAM user sign in** form, which is for users an administrator created inside an account. Because you own the new account, click **Sign in using root user email** instead.



Screenshot 1. *The AWS sign-in page. Click Sign in using root user email (red box).*

On the next page keep **Root user** selected, type the email you registered with, and click **Next**. AWS then asks for your password and, if you turned it on, your MFA code. Type them only on this real AWS page.



Screenshot 2. *Root user sign-in: keep Root user selected and enter your email, then click Next (red boxes).*

Step 1.6 Secure the account and cap your spending

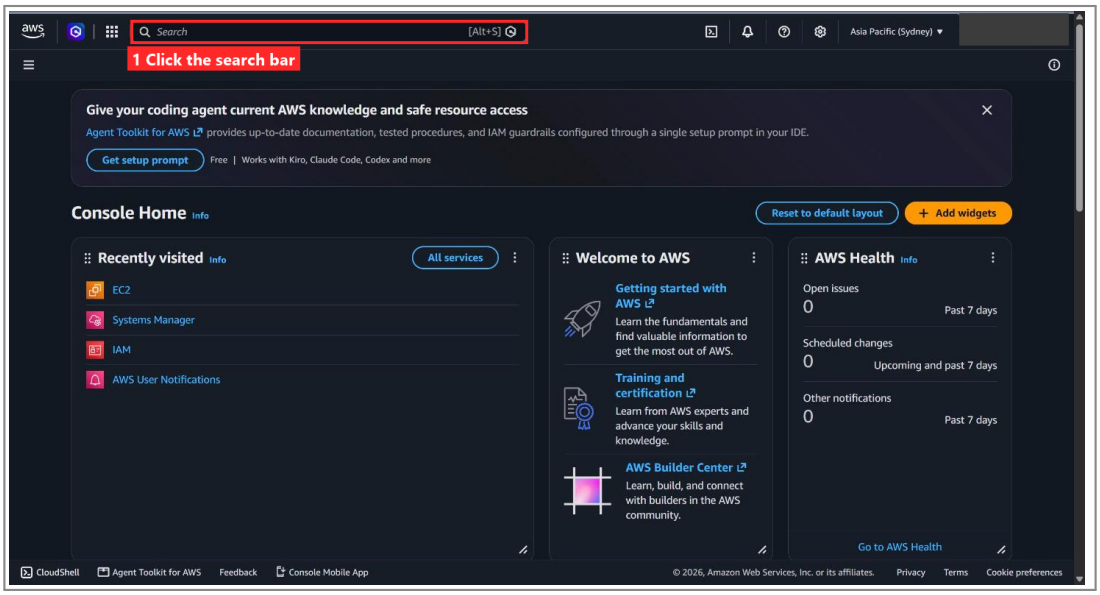
- **Turn on MFA:** open the account menu, choose **Security credentials**, and add a multi-factor authentication device.
- **Create a budget:** go to **Billing > Budgets > Create budget**. Choose a zero-spend or $10 monthly budget and add an email alert, so you hear about unexpected charges early.

2. Launch the EC2 instance

EC2 is the AWS service that provides virtual machines, which AWS calls **instances**. In this part you create one Windows instance of size **m7i-flex.large**, choose a key pair that protects the login, and set a firewall rule so that only your own connection can reach it.

Step 2.1 Open the search bar

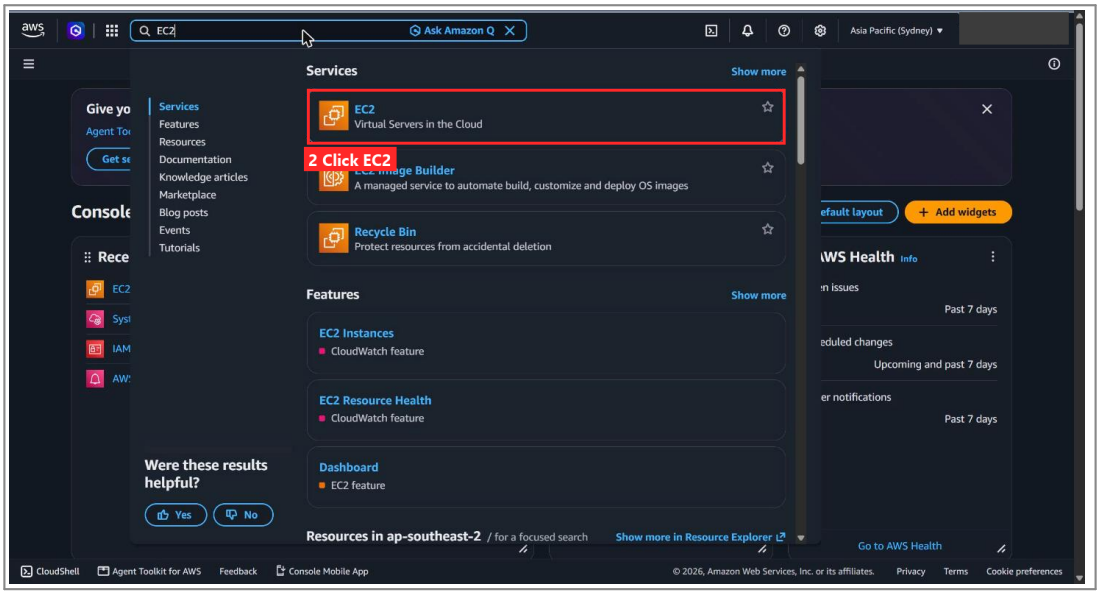
After you sign in you land on **Console Home**. Click the **search bar** at the top of the page.



Screenshot 3. *Console Home after sign-in. Click the search bar (red box).*

Step 2.2 Open EC2

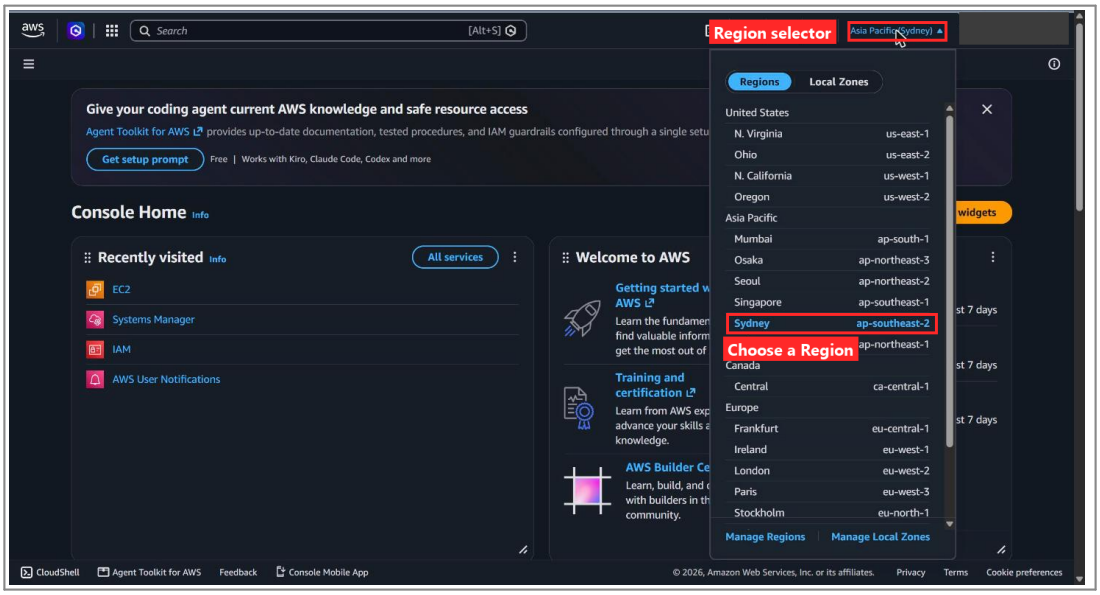
Type **EC2** and click the **EC2 - Virtual Servers in the Cloud** result. If EC2 appears under **Recently visited**, you can click it there instead.



Screenshot 4. *Search results for EC2. Click EC2 (red box).*

Step 2.3 Choose your Region

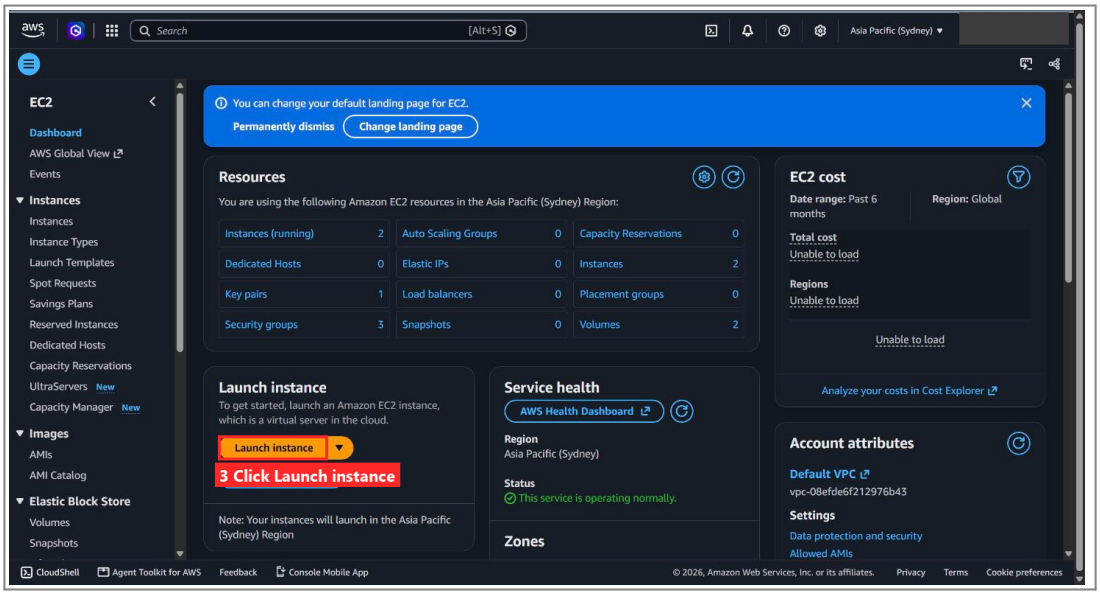
The **Region** is the AWS data centre location, shown at the top right. Pick the one closest to you; it also decides where your VM runs. The screenshots use **Asia Pacific (Sydney) ap-southeast-2**; Singapore (ap-southeast-1) also works. Make sure m7i-flex.large is offered in the Region you choose.



Screenshot 5. *Region selector open; the chosen Region is highlighted.*

Step 2.4 Start the launch

On the **EC2 Dashboard**, click the orange **Launch instance** button.



Screenshot 6. *EC2 Dashboard. Click Launch instance (red box).*

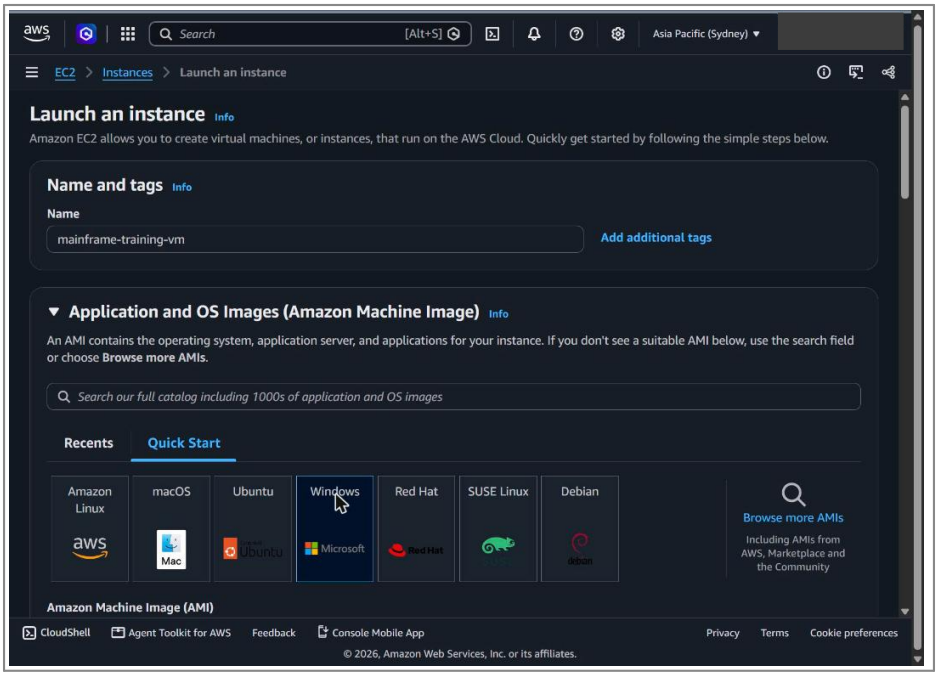
Step 2.5 Fill in the launch form

The launch page is one long form. These are the settings to use; the screenshots that follow show each one.

Field	Value
Name	mainframe-training-vm

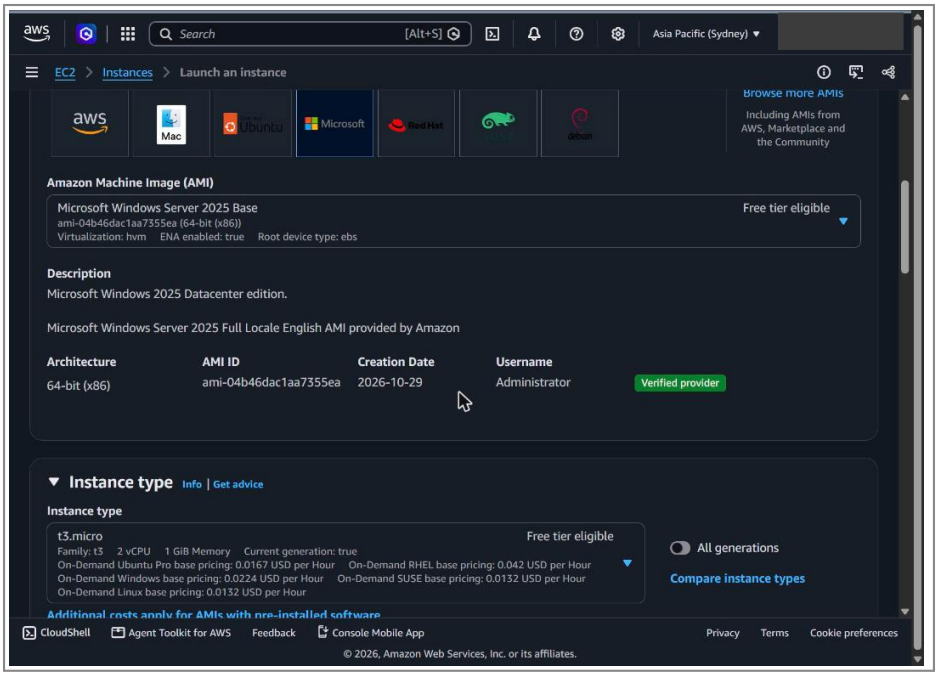
Operating system (AMI)	Microsoft Windows Server 2025 Base (or 2022 Base)
Instance type	m7i-flex.large
Key pair	Create new: mainframe-training-key, RSA, .pem
Network / security group	Create a security group; allow RDP (3389) from My IP, not 0.0.0.0/0
Storage	60 GiB, gp3

Name and operating system. Type the instance name, then choose the **Windows** tile under Quick Start. The name is only a label for you.



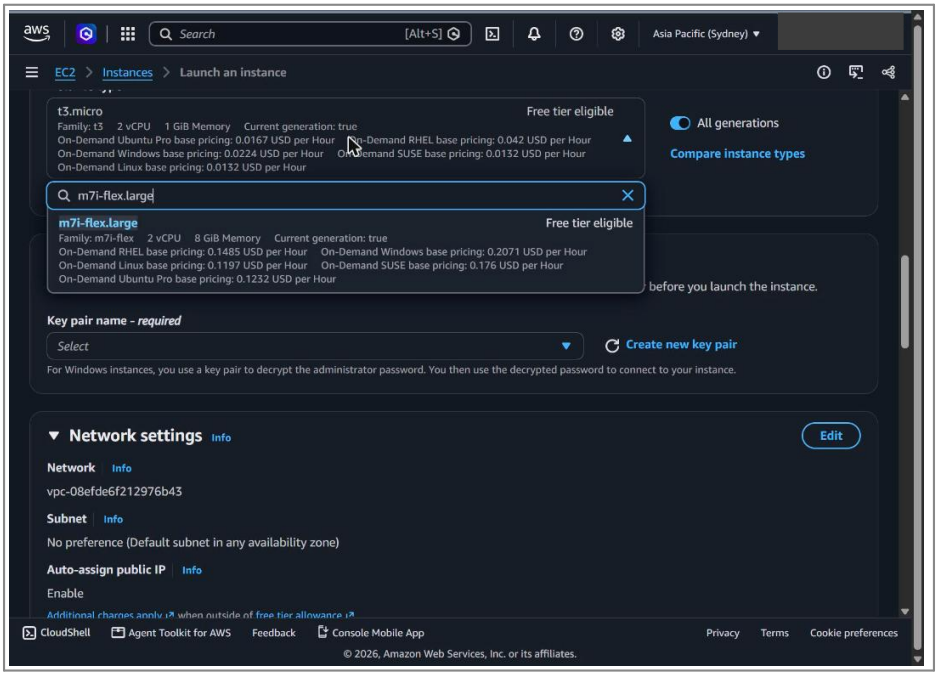
Screenshot 7. *Name entered and the Windows tile selected.*

The image should read **Microsoft Windows Server 2025 Base**, with the default username **Administrator**.

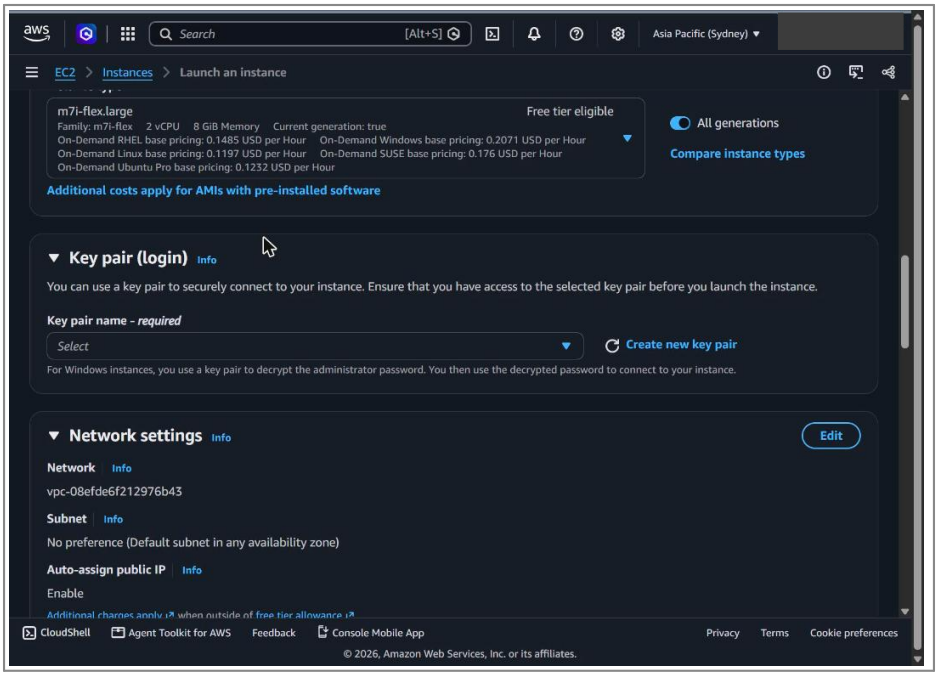


Screenshot 8. *Windows Server 2025 Base selected as the AMI.*

Instance type. Turn on **All generations**, open the Instance type list, and search for **m7i-flex.large** (2 vCPU, 8 GiB). Click it to select it.



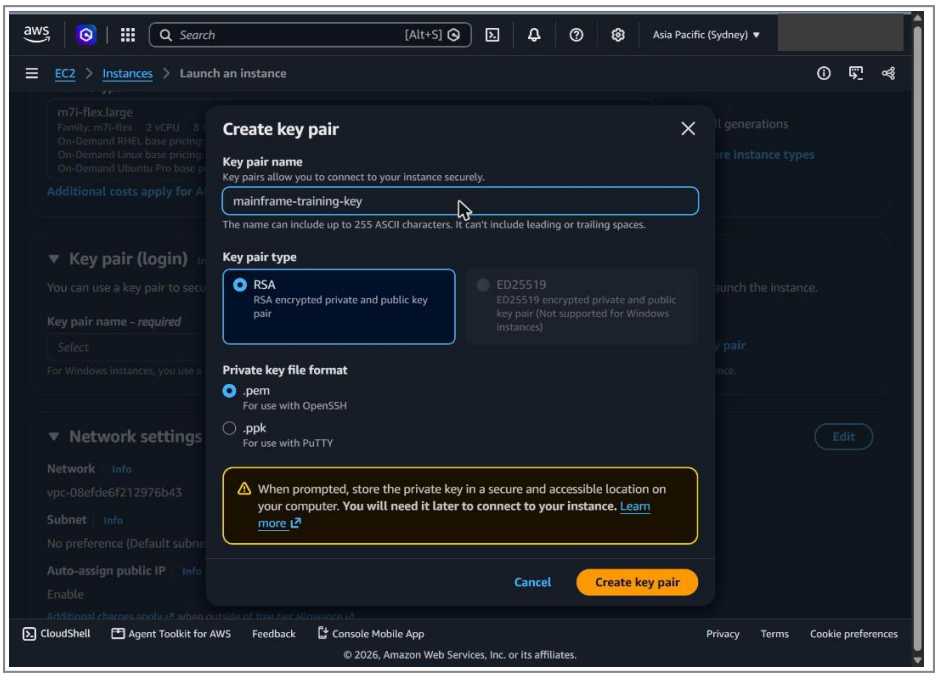
Screenshot 9. *Searching for m7i-flex.large in the Instance type list.*



Screenshot 10. *m7i-flex.large selected. The Key pair (login) section comes next.*

Key pair. A key pair is a private file that proves you may log in. Click **Create new key pair**, name it *mainframe-training-key*, keep **RSA** and **.pem**, and click **Create key pair**.

> The **.pem** file downloads only once and cannot be downloaded again. Save it somewhere safe and never share it. You need it in Part 3 to get the Windows password.



Screenshot 11. *Create key pair dialog with the name entered, RSA and .pem selected.*

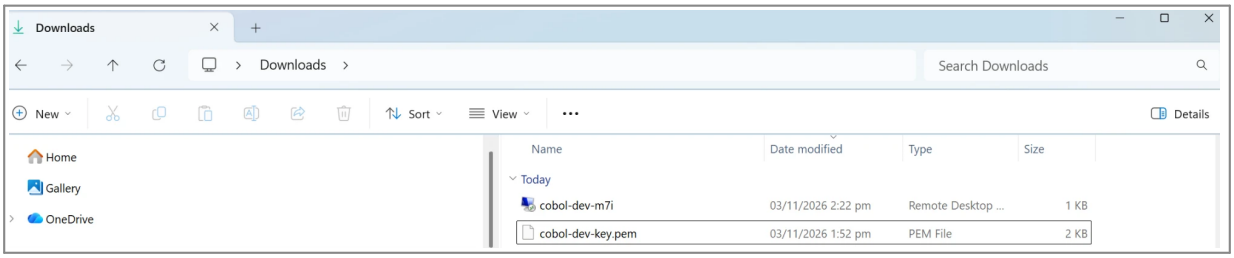
When you click **Create key pair**, a green banner confirms that the key pair was created, and your browser downloads the **.pem** file at the same moment.



Screenshot 12. *The green banner confirming Successfully created key pair.*

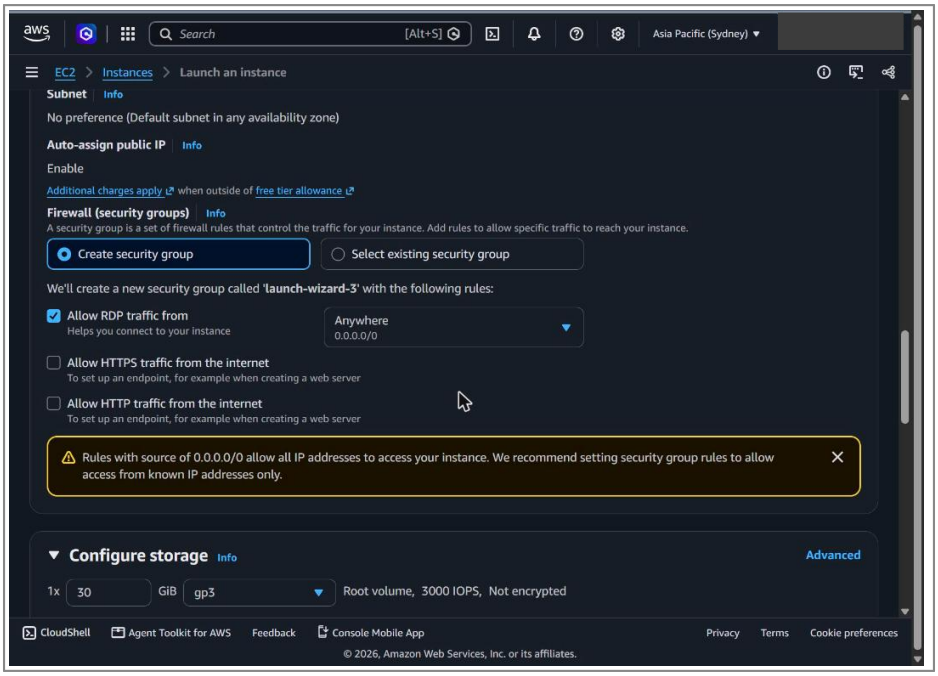
> **Look for the download popup.** Right after you click **Create key pair**, a small popup appears at the top right of your browser to show that the file was downloaded. It disappears after a few seconds. You can click the download icon in the toolbar to see it again.

Then open your **Downloads** folder and confirm the file is there, and move it somewhere safe, such as a folder only you can open.



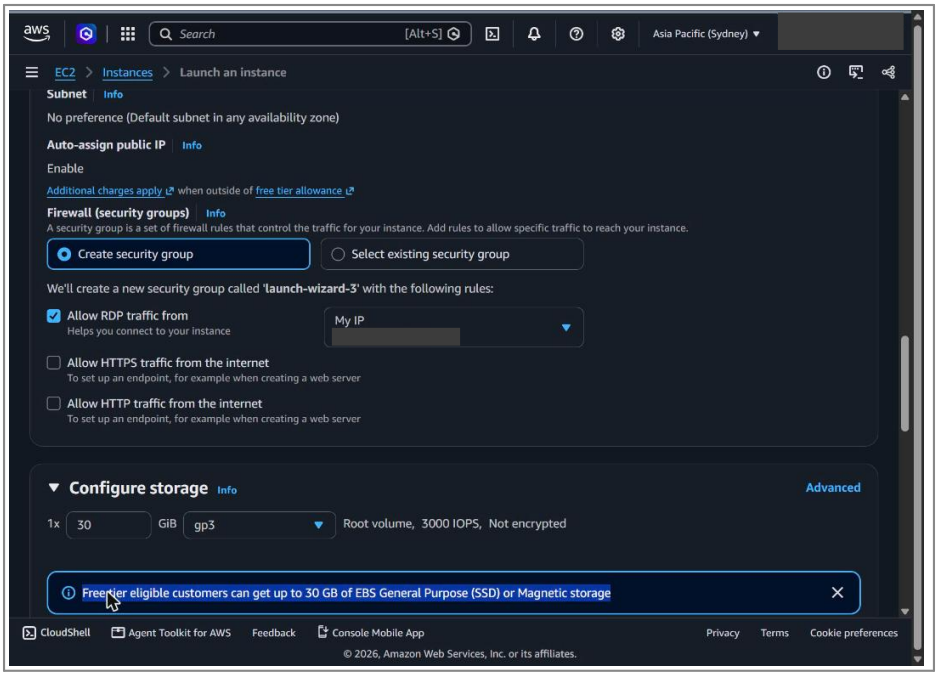
Screenshot 13. *The Downloads folder after the key pair is created: the .pem file (highlighted) is saved here. The example key is named cobol-dev-key.pem; yours will be mainframe-training-key.pem. The .rdp file beside it is the shortcut downloaded in Part 3.*

Network settings (firewall). Keep **Create security group** with **Allow RDP traffic from** ticked. The default source is **Anywhere (0.0.0.0/0)**, which opens RDP to the whole internet, and AWS warns you about it.



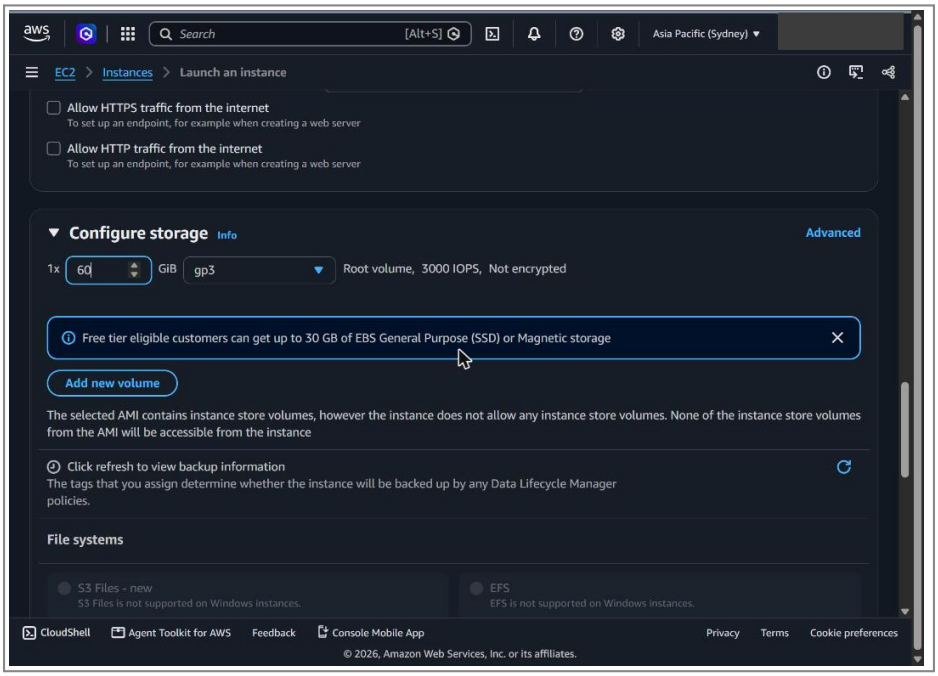
Screenshot 14. *Default RDP rule: Anywhere (0.0.0.0/0). Not recommended.*

Change the source to **My IP**, so only your current public IP address can reach the VM.



Screenshot 15. *RDP source changed to My IP (your IP address is hidden here).*

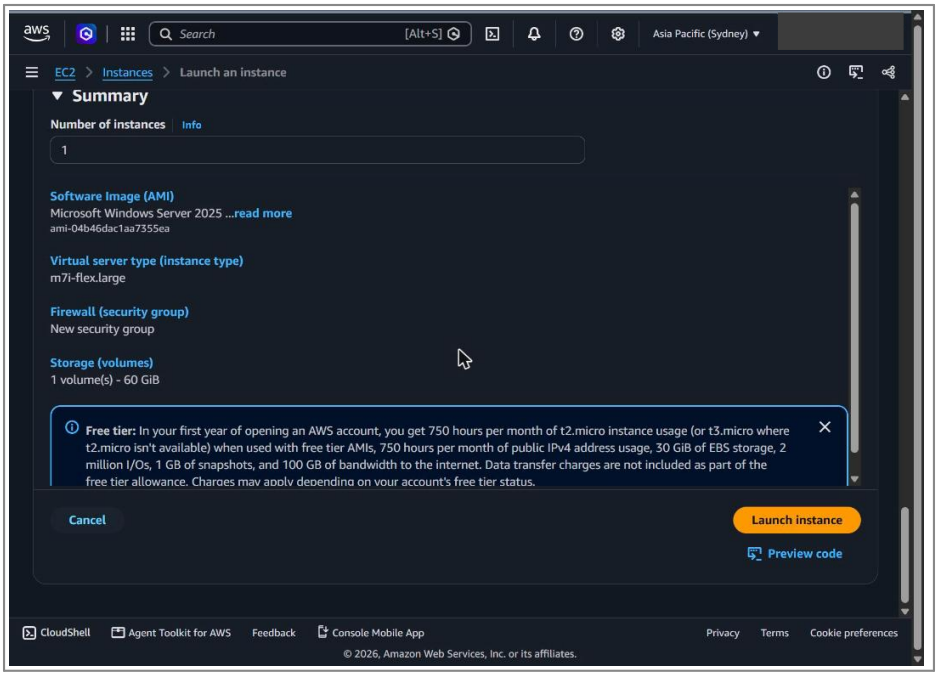
Storage. Set the root volume to **60 GiB**, type gp3. This leaves room for the tools and your lab files.



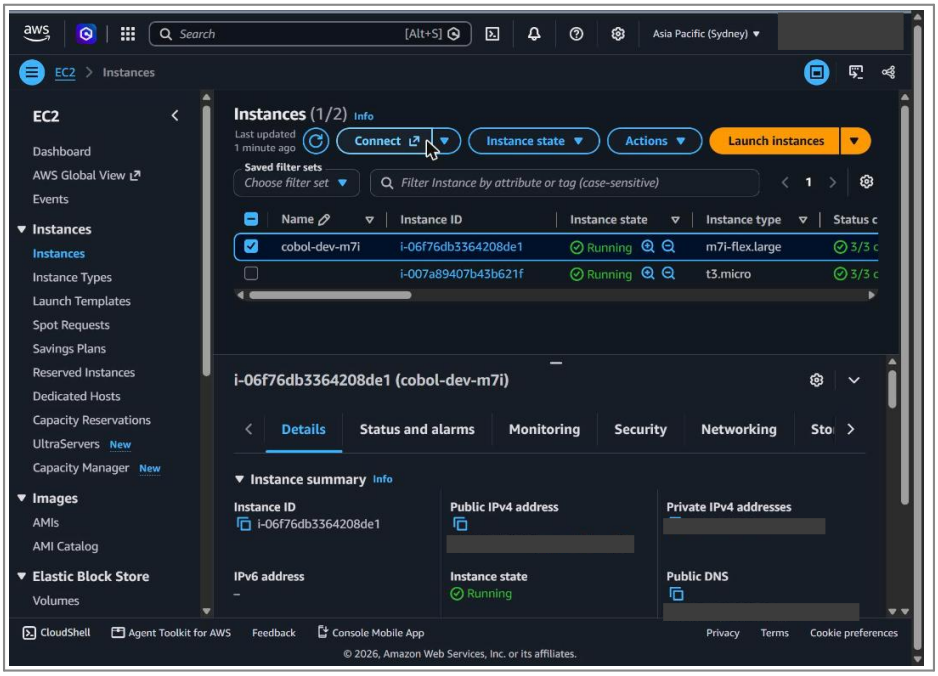
Screenshot 16. *Configure storage: 60 GiB gp3.*

Step 2.6 Launch and wait until it is running

Check the **Summary** panel on the right (image, instance type, security group, 60 GiB) and click **Launch instance**. Then open the **Instances** list and wait until **Instance state** shows **Running** and the **Status checks** pass.



Screenshot 17. *Summary panel with the Launch instance button (m7i-flex.large, one 60 GiB volume).*



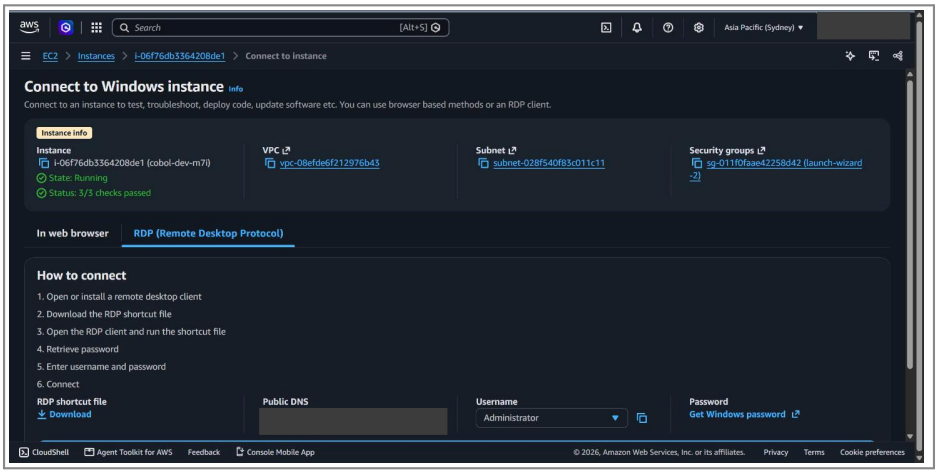
Screenshot 18. *Instances list: m7i-flex.large Running with status checks passed. The example instance here is named cobol-dev-m7i; yours will be named mainframe-training-vm.*

3. Connect with Remote Desktop

Remote Desktop Protocol (**RDP**) lets you see and control the VM's Windows desktop from your own PC. To log in you need the instance's **Public DNS** (its internet address), the user name **Administrator**, and a password that you decrypt with your .pem file.

Step 3.1 Open the connection page

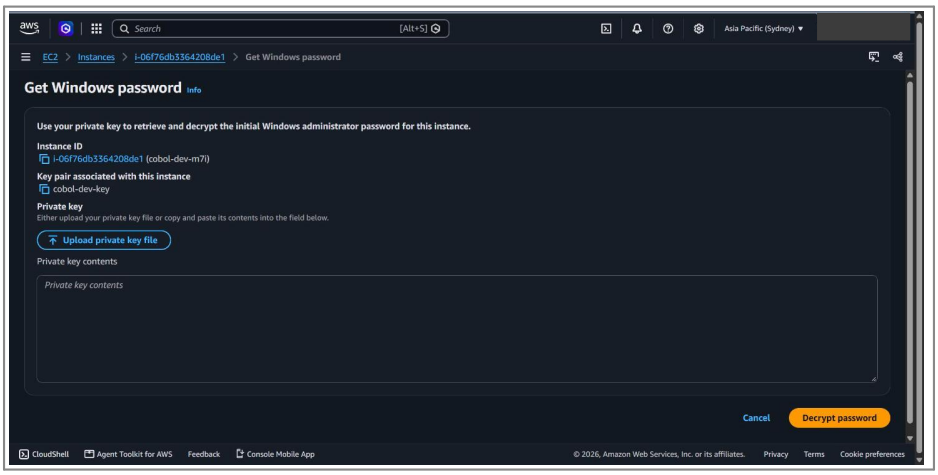
In the Instances list, select your instance and click **Connect**. Open the **RDP (Remote Desktop Protocol)** tab. Note the **Public DNS** and the user name **Administrator**. The page also offers an **RDP shortcut file** download, which is the easiest way to connect (Step 3.3).



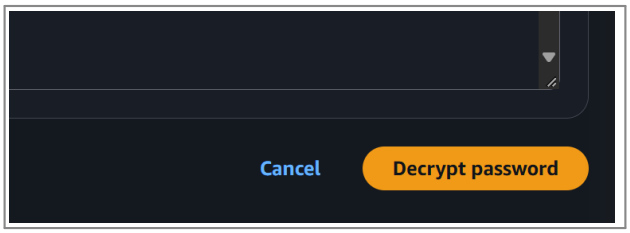
Screenshot 19. *Connect to Windows instance, RDP tab: Public DNS (hidden), Administrator, Get Windows password.*

Step 3.2 Decrypt the Administrator password

Click **Get Windows password**. On the next page click **Upload private key file** and choose your **.pem** file (or paste its contents), then click **Decrypt password**. Copy the password straight away.



Screenshot 20. *Get Windows password page: upload the private key file.*



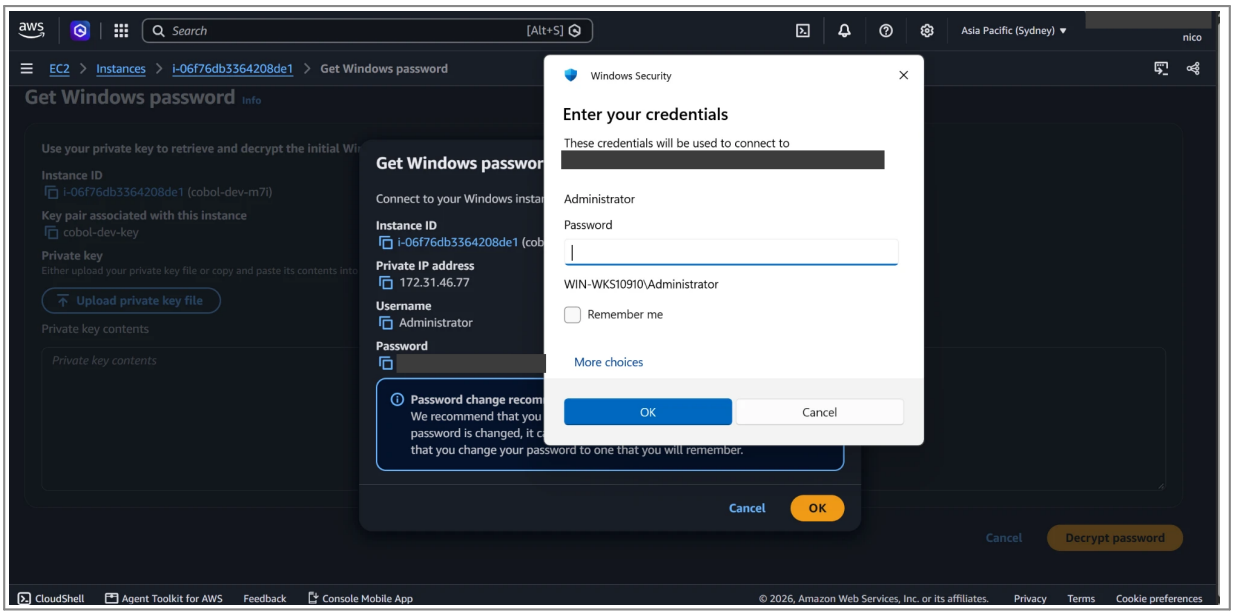
Screenshot 21. *After the key is loaded, click Decrypt password.*

> Never screenshot or share this page once the key or the password is showing, and keep the password out of chat and email.

Step 3.3 Log in

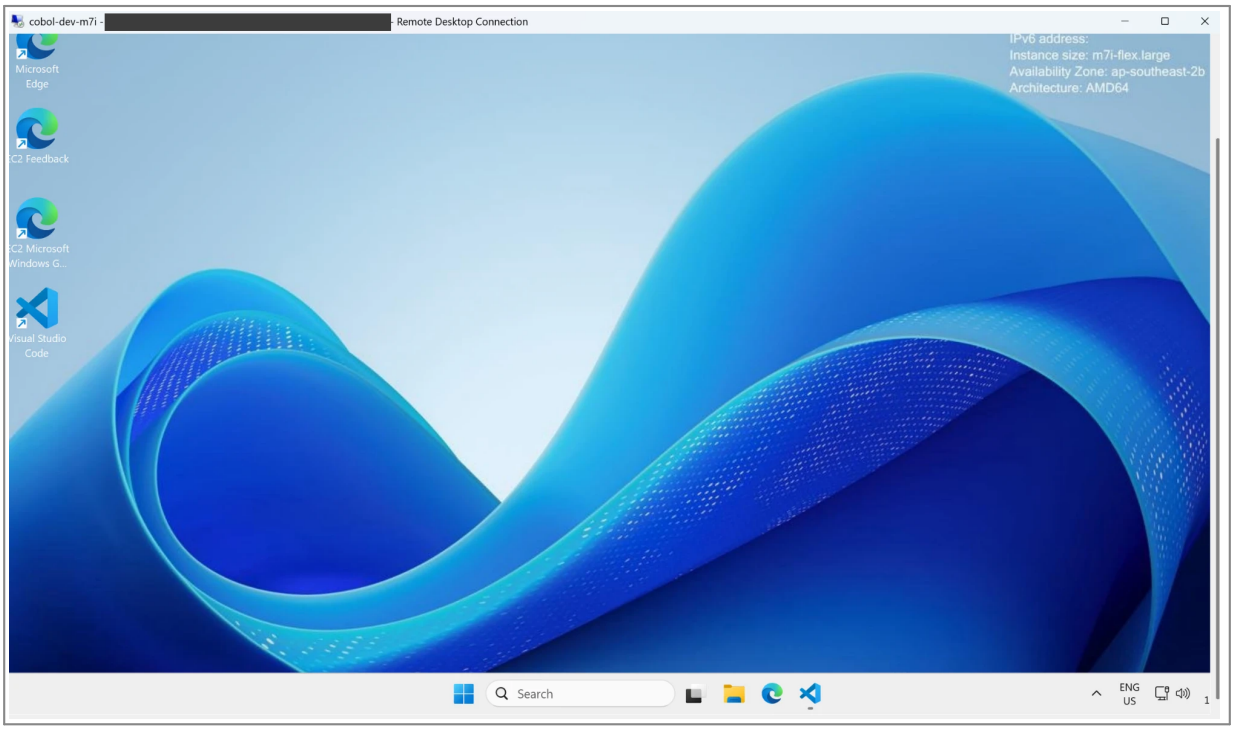
Option A (easiest): download the **RDP shortcut file** from the Connect page and open it. **Option B:** press **Win+R**, type `mstsc`, press Enter, enter the Public DNS as the computer and `Administrator` as the user, and click **Connect**. On a Mac, use the Microsoft Remote Desktop app.

Windows shows a **Windows Security** prompt for the Administrator account. Paste the decrypted password, click **OK**, and accept the certificate warning.



Screenshot 22. *The Windows Security prompt for Administrator. Paste the decrypted password (hidden here) and click OK.*

When the connection succeeds you see the Windows Server desktop inside a Remote Desktop window.



Screenshot 23. *Connected: the Windows Server desktop inside the Remote Desktop window. The instance size, m7i-flex.large, is shown at the top right.*

Opening a terminal: Command Prompt or PowerShell. Either one works for most commands in this guide. Shortcuts:
- Press **Win+R**, type `cmd` (Command Prompt) or `powershell` (PowerShell), then press **Enter**.
- To open it **as Administrator**, press **Win+X** and then **A**, or type the name in the Run box and press **Ctrl+Shift+Enter**.
- Inside a window, type `cmd` to switch to Command Prompt, or `powershell` to switch to PowerShell.

The time-zone command in Step 3.4 and the Chocolatey, PATH and npm-folder commands in Part 4 are PowerShell commands. The rest also work in Command Prompt.

Optional, from Command Prompt. You can also store the login and connect from the command line (replace the DNS and the password). This keeps the password in your command history, so use it only on your own PC.

```
cmdkey /generic:TERMSRV/ec2-x-x-x-x.ap-southeast-2.compute.amazonaws.com /user:Administrator /pass:<password>
mstsc /v:ec2-x-x-x-x.ap-southeast-2.compute.amazonaws.com
```

Step 3.4 First actions on the VM

Open **Windows PowerShell as Administrator** on the VM (press **Win+X**, then **A**). Replace the long generated password with one you will remember, and set your time zone (the example uses Singapore; run `Get-TimeZone -ListAvailable` to see the others):

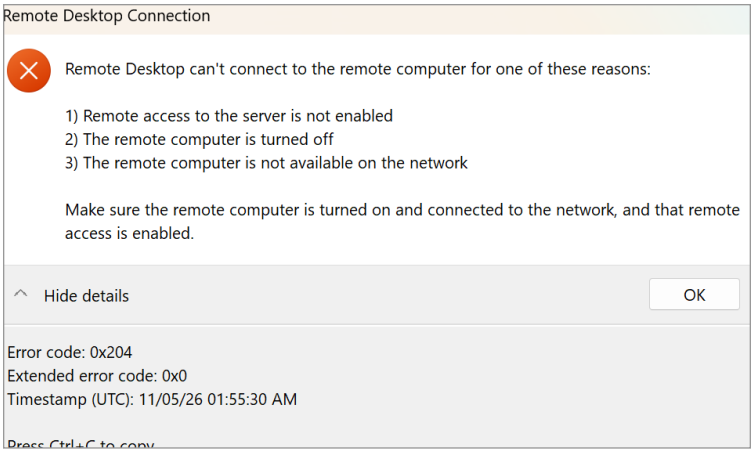
```
net user Administrator *
Set-TimeZone -Id "Singapore Standard Time"
```

Troubleshooting

What you see	Likely cause	What to do
Connection timed out	Your public IP changed	Edit the security group rule and set the source to My IP again
Password not available	The VM is still starting	Wait about 4 minutes after launch and try again
Credentials rejected	Wrong or old password	Decrypt the password again with the same .pem file
RDP error 0x204	Your public IP changed	Set the RDP rule source to My IP; use the new Public IP

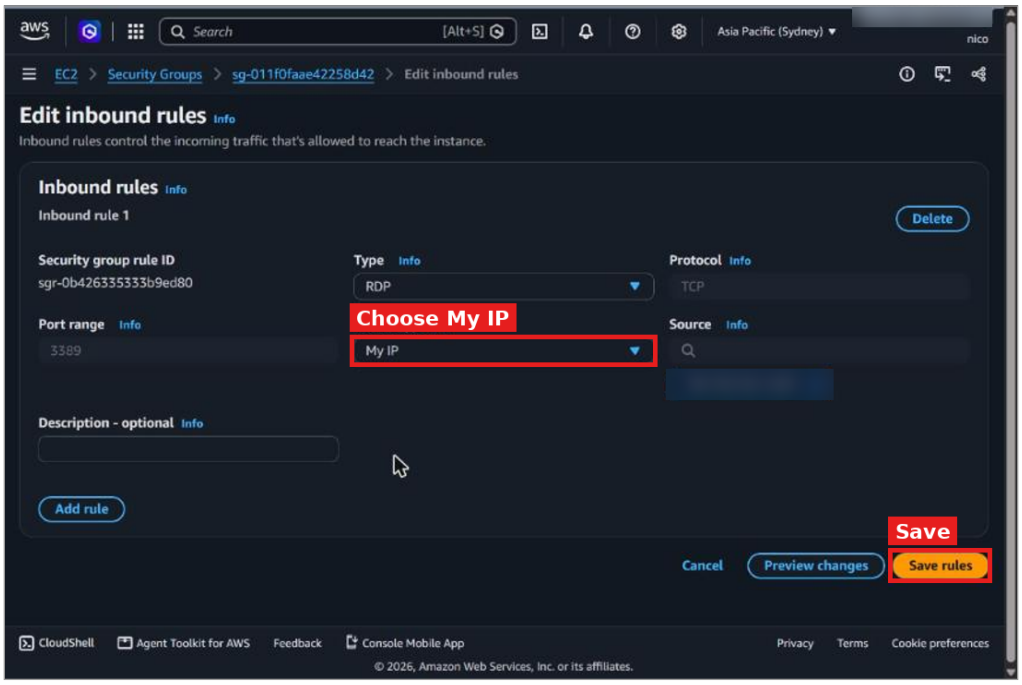
If Remote Desktop shows error 0x204

Your PC is blocked by the security group, usually because your public IP changed since you set the rule.



Screenshot 23a. The Remote Desktop error 0x204 you will see.

Fix: EC2 > Instances > your instance > Security tab > click the sg- link > Edit inbound rules. On the RDP rule set Source type to My IP, click Save rules, then connect again to the instance's current Public IPv4 address.



Screenshot 23b. Edit inbound rules: set Source type to My IP, then Save rules (your IP is hidden here).

4. Install the tools

In this part you install the six tools, one command each. All screenshots here were taken on the training VM while the commands ran. Open **Windows PowerShell as Administrator** on the VM (press **Win+X** then **A**, or right-click Start and choose **Terminal (Admin)**).

> Windows Server does not include `winget`, so these steps use **Chocolatey**, a package installer where each tool is one command, such as `choco install git -y`. The `-y` flag answers the confirmation prompts for you.

Step 4.1 Install Chocolatey (one time)

Skip this step if Chocolatey is already installed.

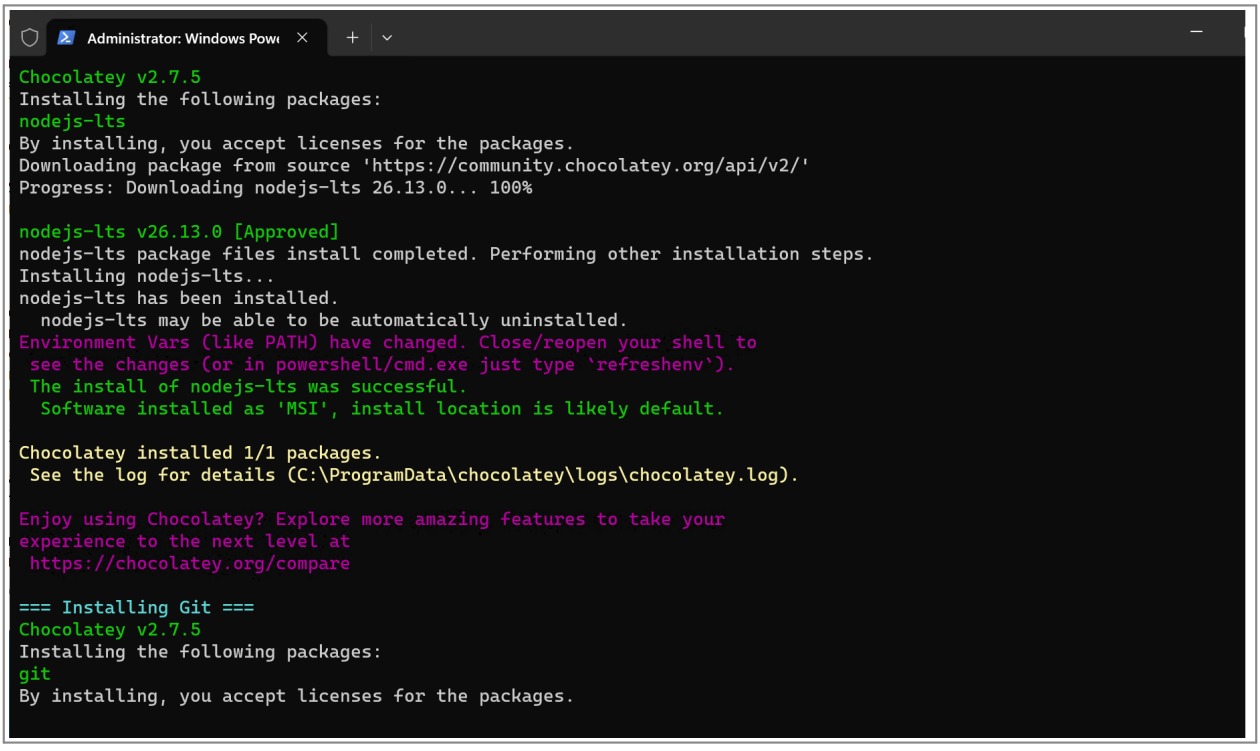
```
Set-ExecutionPolicy Bypass -Scope Process -Force;
[Net.ServicePointManager]::SecurityProtocol = [Net.ServicePointManager]::SecurityProtocol -bor 3072;
Invoke-Expression ((New-Object Net.WebClient).DownloadString('https://community.chocolatey.org/install.ps1'))
```

If you see "A positional parameter cannot be found": the semicolons were lost. Add ; after -Force and after 3072, then run it again.

Step 4.2 Node.js (LTS)

Node.js runs JavaScript tools and provides `npm`, which Claude Code needs.

```
choco install nodejs-lts -y
```

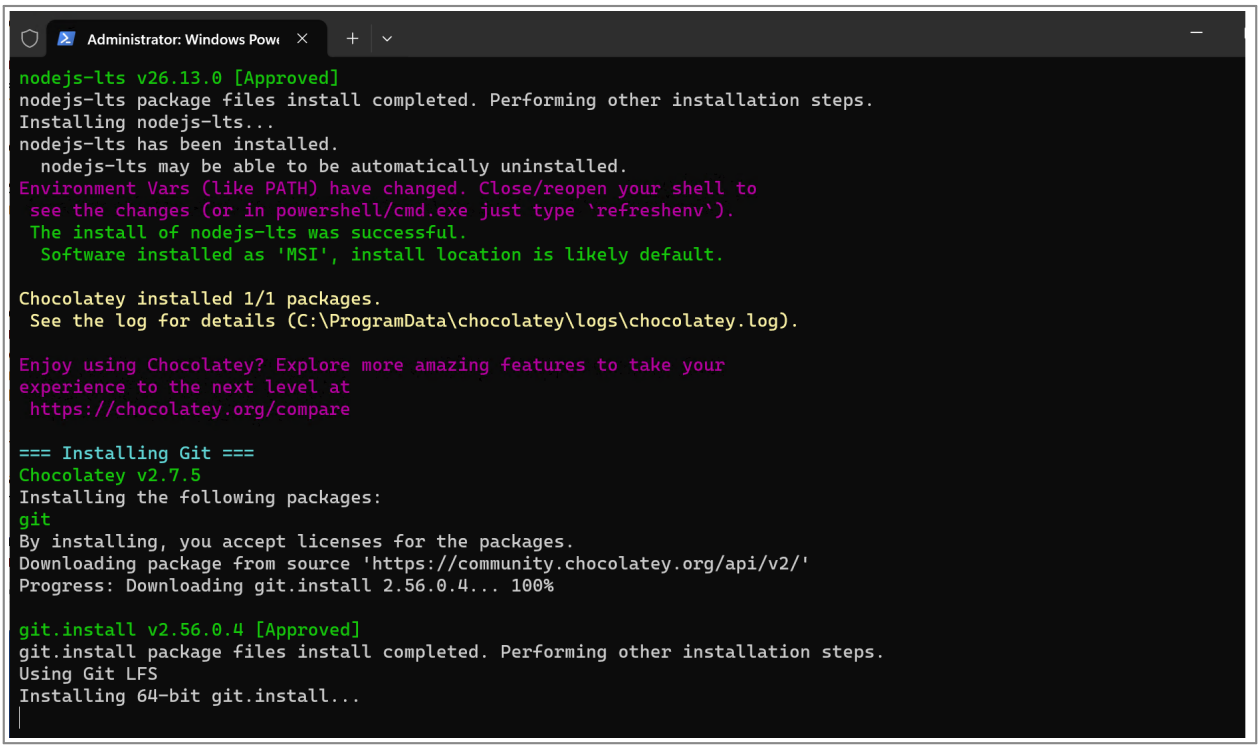


Screenshot 24. *Node.js LTS installed (nodejs-lts 26.13.0). Git comes next.*

Step 4.3 Git

Git is the version control tool used throughout the labs.

```
choco install git -y
```

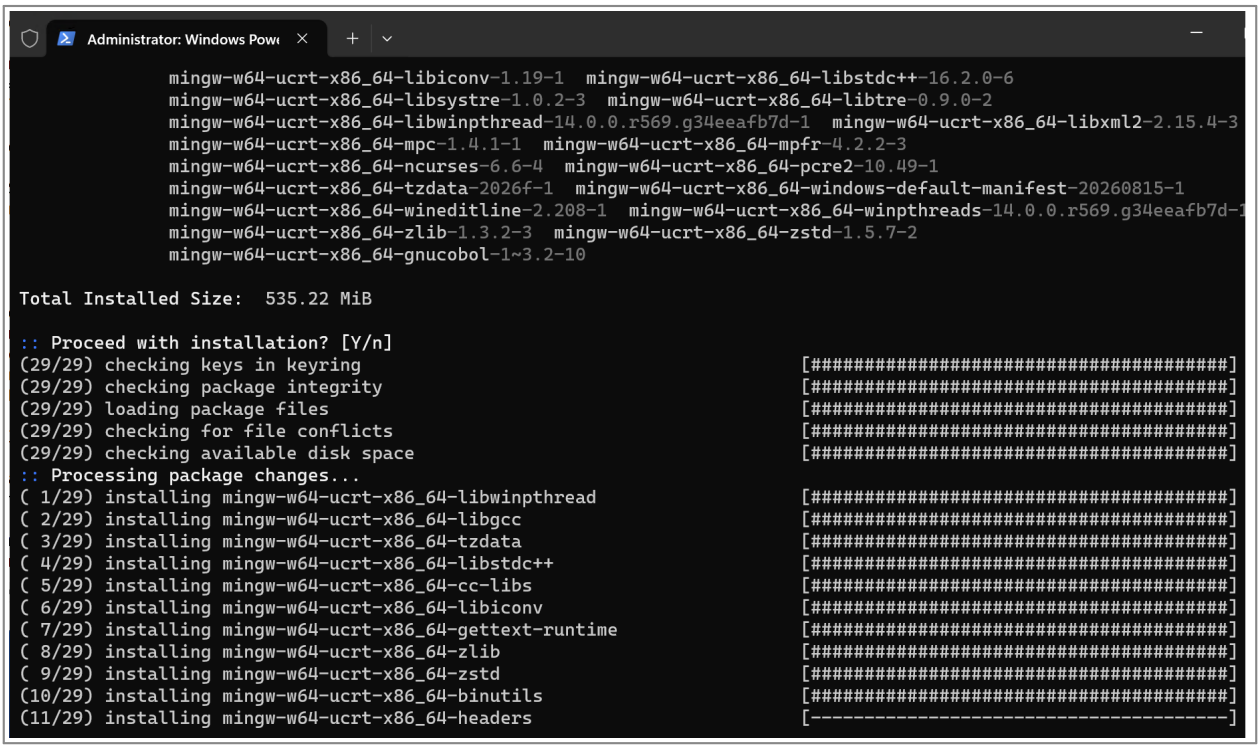


Screenshot 25. *Git installed (2.56.0.4). The MSYS2 and GnuCOBOL step begins.*

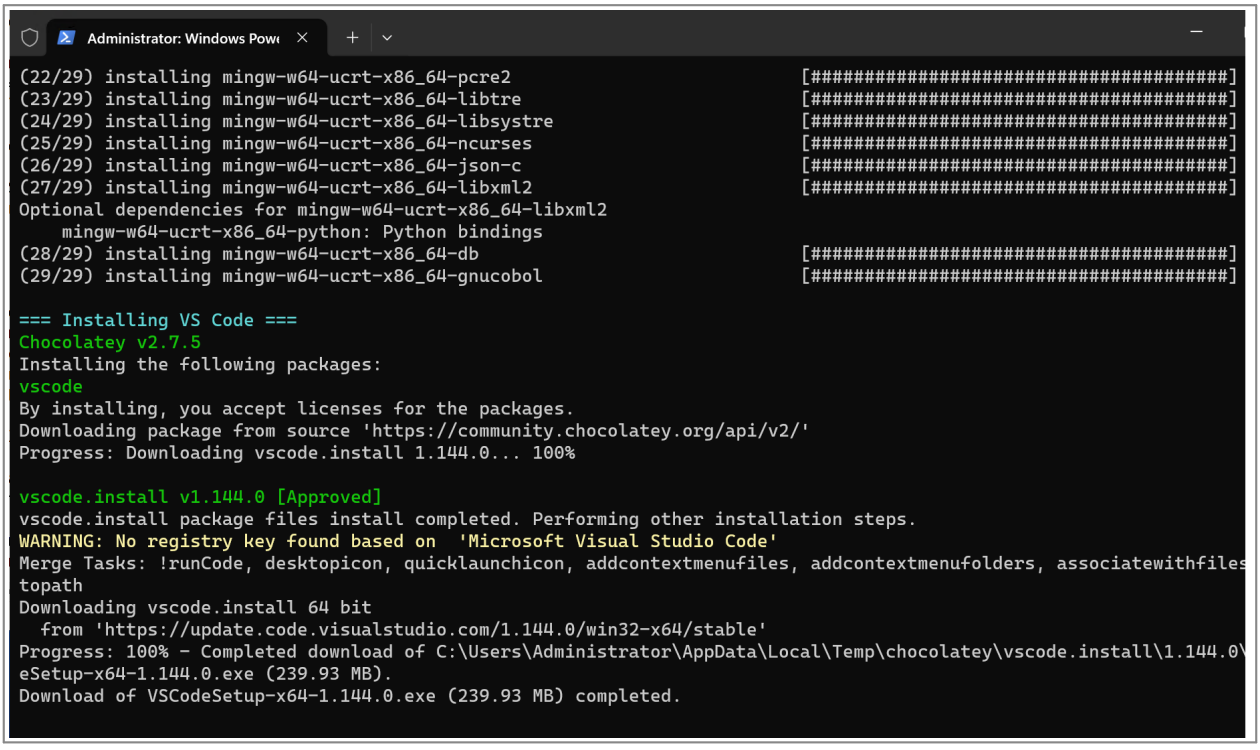
Step 4.4 GnuCOBOL

GnuCOBOL is the COBOL compiler (`cobc`). It comes through MSYS2, a toolkit for building software on Windows. First install MSYS2, then install the GnuCOBOL package with `pacman`, and add the compiler folder to the machine PATH so `cobc` works from any terminal.

```
choco install msys2 -y
& "C:\tools\msys64\usr\bin\bash.exe" -lc 'pacman -Sy --noconfirm --needed mingw-w64-ucrt-x86_64-gnucobol'
$p = [Environment]::GetEnvironmentVariable('Path','Machine')
[Environment]::SetEnvironmentVariable('Path', "$p;C:\tools\msys64\ucrt64\bin", 'Machine')
```



Screenshot 26. *pacman lists the GnuCOBOL package set (29 packages, about 535 MiB) and starts installing it.*

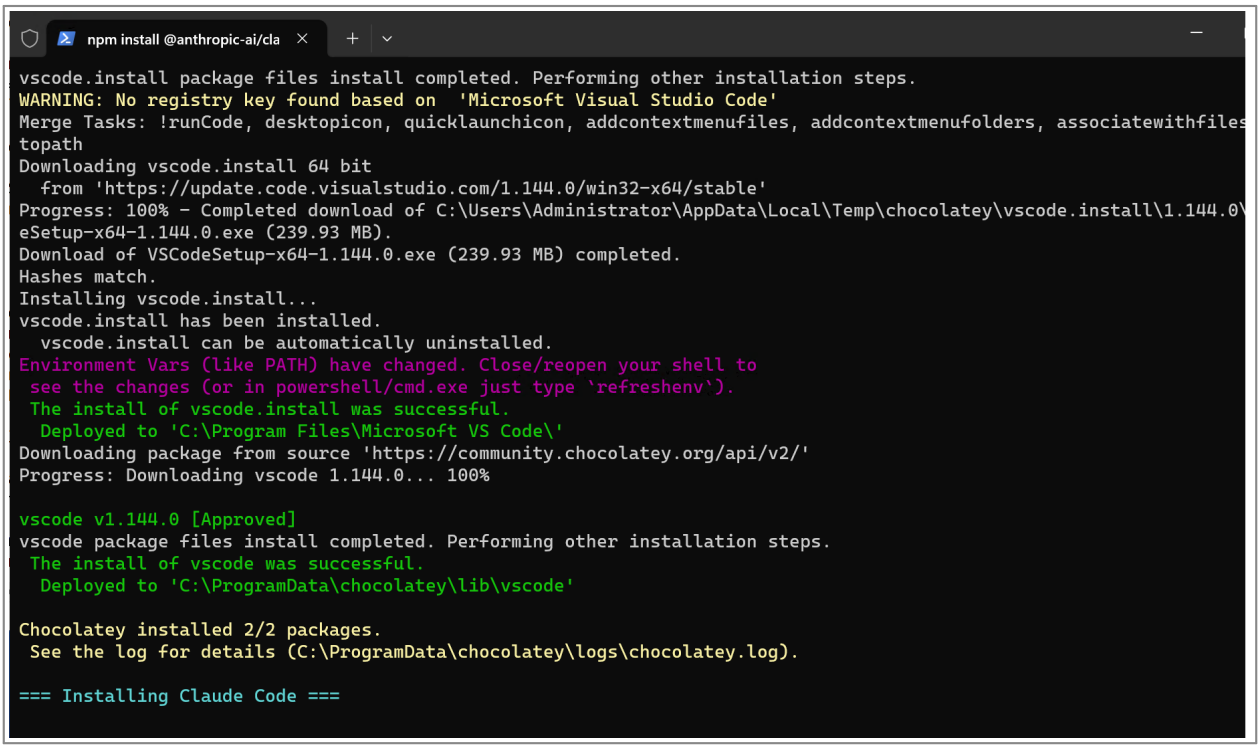


Screenshot 27. *All 29 packages installed, ending with mingw-w64-ucrt-x86_64-gnucobol. VS Code starts next.*

Step 4.5 Visual Studio Code

VS Code is the editor you will use to write and read code.

```
choco install vscode -y
```

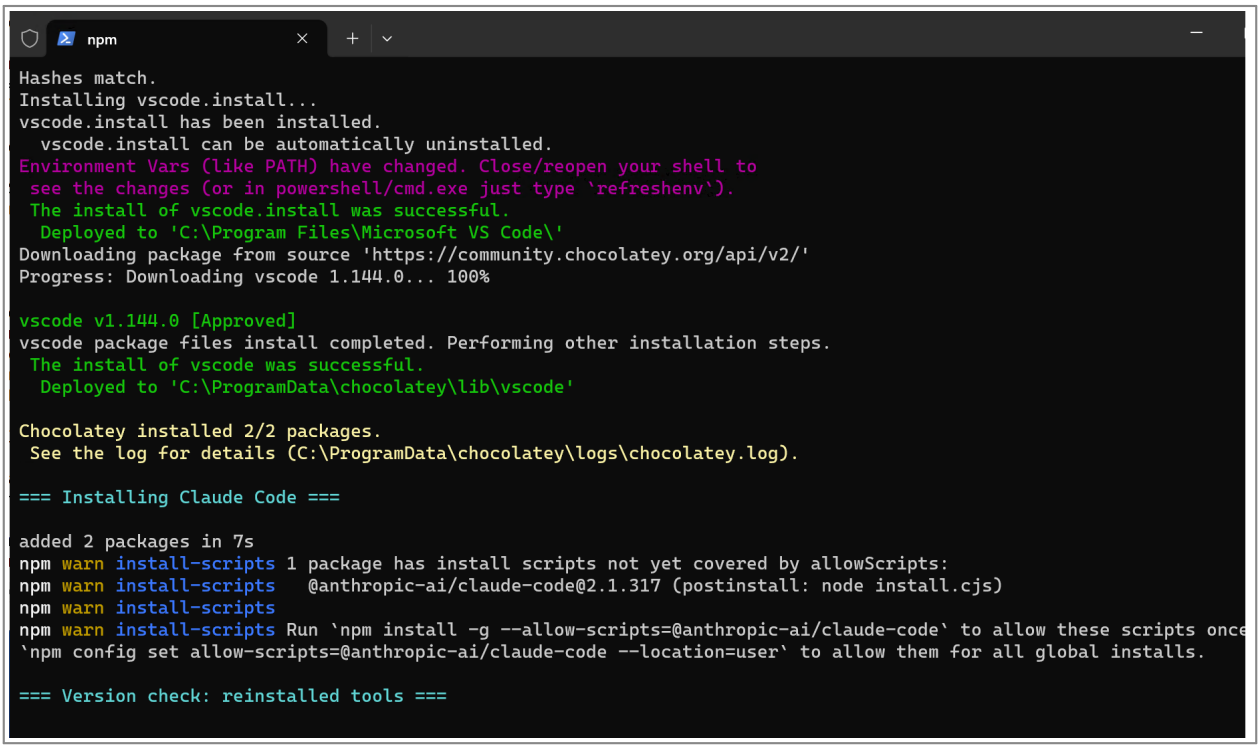


Screenshot 28. *VS Code installed (vscode 1.144.0). Claude Code starts next.*

Step 4.6 Claude Code

Claude Code installs through npm, so it needs Node.js from Step 4.2. First set a global npm folder, then install the package.

```
New-Item -ItemType Directory -Force C:\npm-global | Out-Null
npm config set prefix C:\npm-global
npm install -g @anthropic-ai/claude-code
```



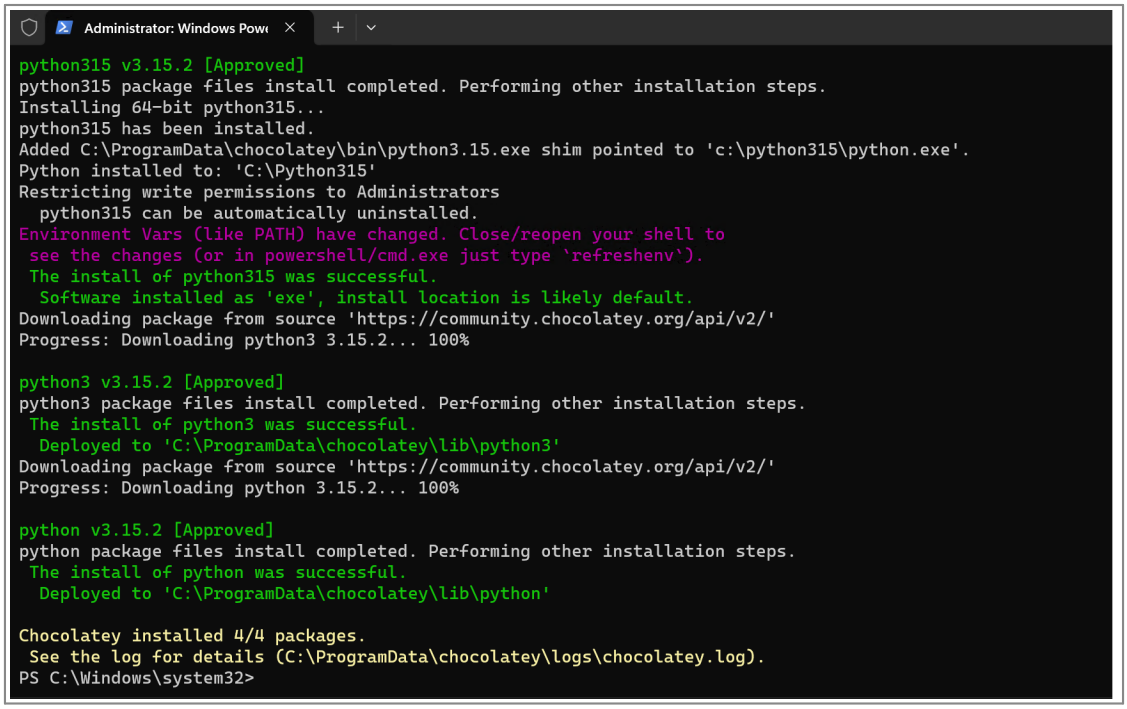
Screenshot 29. *npm installs Claude Code (added 2 packages). The version check starts.*

The first time you run `claude`, it asks you to sign in. Open a **new** terminal first so the updated PATH applies, then follow the browser prompt with your training account.

Step 4.7 Python 3

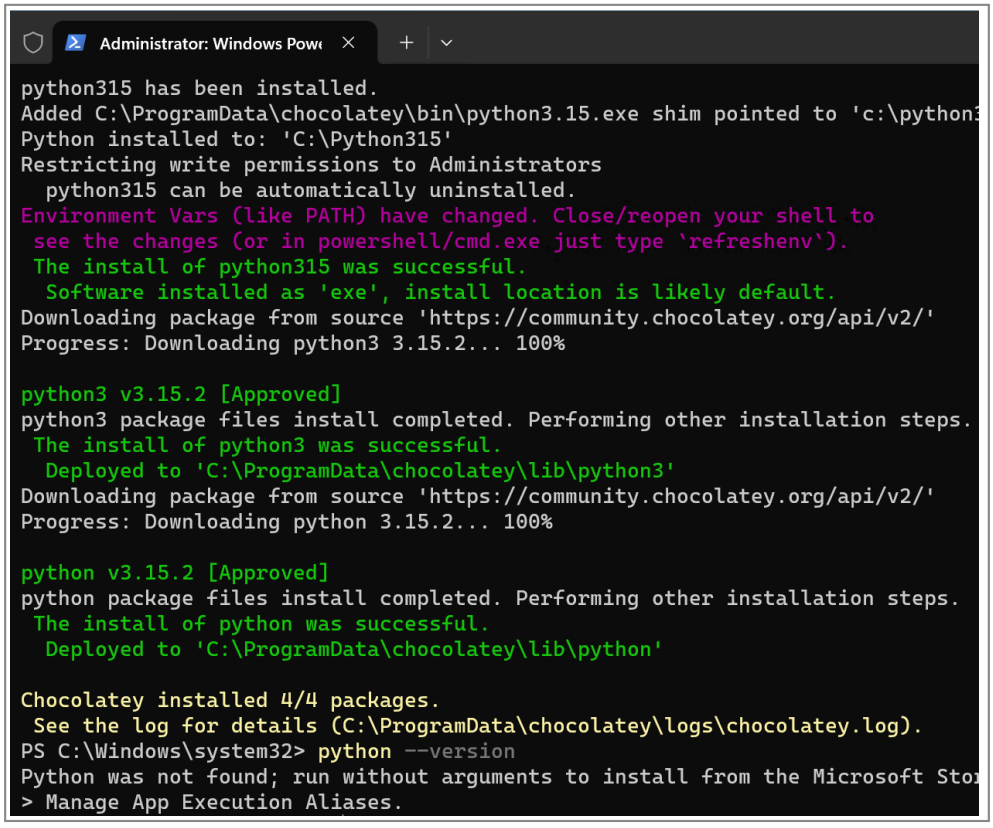
Python is installed the same way. Run this in the Administrator prompt:

```
choco install python -y
```



Screenshot 30. *Python installed through Chocolatey (python 3.15.2 plus its python3 and python315 packages: 4 of 4 installed).*

Open a new terminal before you check the version. In the same window the PATH has not refreshed yet, so `python --version` still says Python was not found. Close the window, open a new Administrator PowerShell, and run it again.



Screenshot 31. *Right after the install, python --version fails in the same window until a new terminal is opened.*

Step 4.8 VS Code extensions (optional)

Add the Python and COBOL language support to VS Code:

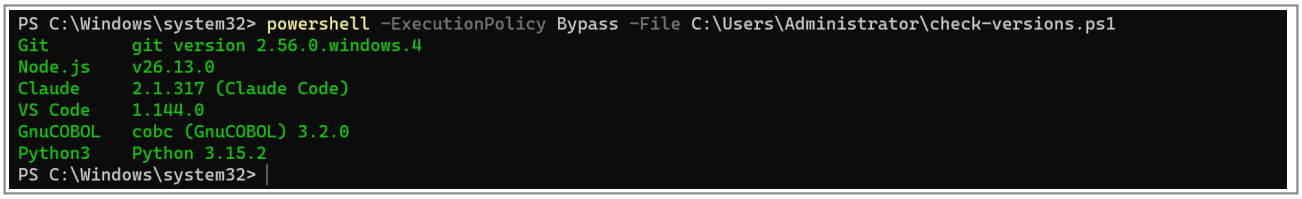
```
code --install-extension ms-python.python
code --install-extension bitlang.cobol
```

5. Verify every tool

Finish by checking that all six tools report a version. Run these in PowerShell or Command Prompt:

```
git --version
cobc --version
python --version
node --version
claude --version
code --version
```

The screenshot shows the same six checks run together on the training VM. Each tool prints its version in green. Your VM may show the same or newer versions, and every line should show a version number.



Screenshot 32. *Version check on the VM: Git, Node.js, Claude Code, VS Code, GnuCOBOL and Python 3 all report a version.*

6. Daily care and cleanup

Stop the VM when you are not using it. In the EC2 console choose **Instance state > Stop**. Compute billing stops, but the 60 GiB disk still bills. After a restart the public IP address changes, so read the new Public DNS from the Connect page. When the training ends, choose **Terminate instance** and delete any leftover volumes.

Final checklist

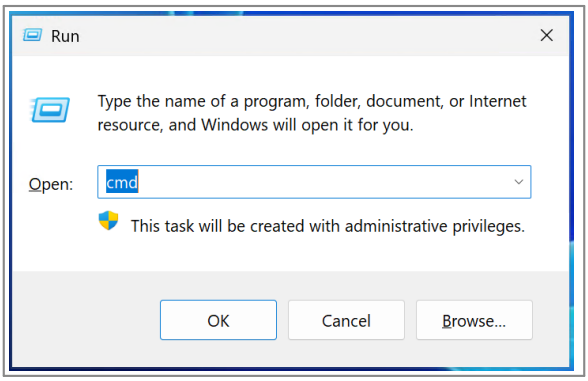
- AWS account created, MFA on, budget alert set
- Instance m7i-flex.large running, and Remote Desktop works
- Git, GnuCOBOL, Python 3, Node.js, Claude Code and VS Code all return versions

7. Bonus (optional): start Claude Code and sign in

If you finished Parts 1 to 6, you can go one step further and open Claude Code in the terminal and sign in with your own account. This is optional, but it is the tool you will use in the training labs, so it is worth confirming now.

Step 7.1 Open a terminal

Press **Win+R** on the VM, type `cmd` (Command Prompt) or `powershell` (PowerShell), and press Enter. The Run box notes that the task will be created with administrative privileges, which is fine here. Either window works for Claude Code. Always open a **new** window after installing tools: only a new window picks up the updated PATH, which is what lets it find the `claude` command.



Screenshot 33. *The Run box with cmd typed in. Click OK.*

Step 7.2 Start Claude Code

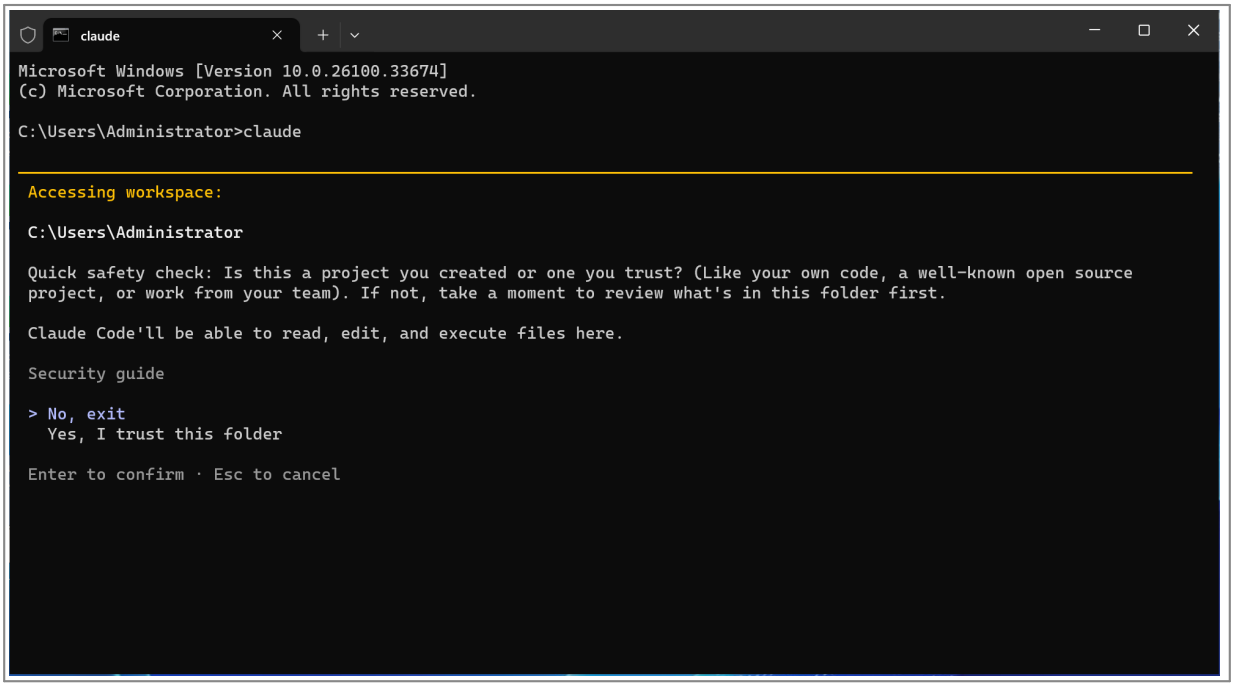
Claude Code works inside the folder where you start it. You can start it right away, or first move to a working folder of your choice. The example uses the `C:\work` folder. Type `claude` and press Enter.

```
mkdir C:\work
cd C:\work
claude
```

Use the folder you prefer. `C:\work` is only an example. Replace it with any folder you like, for example the one where you keep your scripts and projects: `cd D:\scripts` in PowerShell, or `cd /d D:\scripts` in Command Prompt (the `/d` switch also changes the drive). Choose a folder you trust, because Claude Code can read, edit and run files in the folder where you start it.

Step 7.3 Answer the folder question

Claude Code first asks whether you trust the folder, because it can read, edit and run files there. Choose **Yes, I trust this folder** for your own working folder, then press Enter. (Pick **No, exit** for any folder you do not know.)



Screenshot 34. *The folder safety check. Use the arrow keys to choose Yes, I trust this folder, then press Enter.*

Step 7.4 Sign in with your account

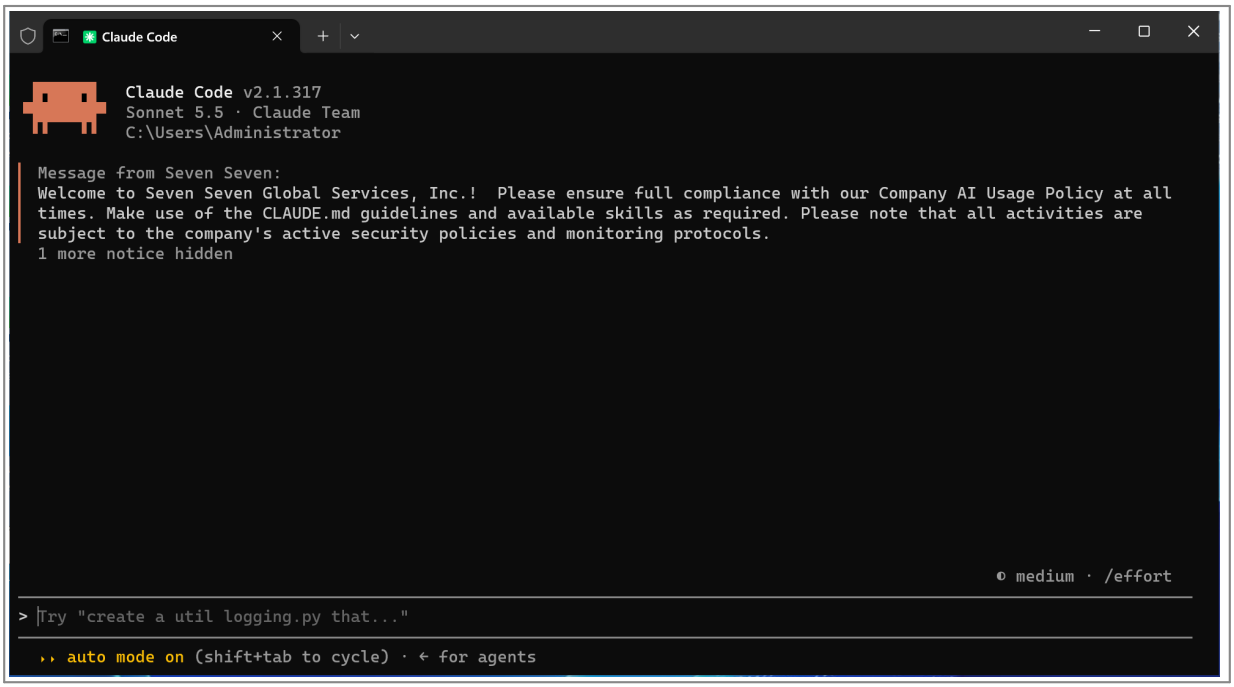
If this is your first run, Claude Code asks you to sign in before it shows the screen above. Choose your **Claude account**, the one your company gave you for the training. Claude Code opens a sign-in page in the browser on the VM (Microsoft Edge). If it does not open, copy the link shown in the terminal and paste it into Edge. Sign in, approve the request, and return to the terminal.

> The training VM in the screenshots was already signed in, so its screens start at the folder question and go straight to the welcome screen. On a fresh VM you will see the sign-in prompt first.

> Sign-in links and codes are private. Never share them and never put them in a screenshot. Type your password only on the real sign-in page, never into the terminal.

Step 7.5 Check that it works

When you are signed in, Claude Code shows its welcome screen with the version, your plan and the current folder, followed by the prompt. Type a short question, such as `what folder am I in?`, and press Enter. A reply means everything works. Type `/exit` to leave Claude Code.



Screenshot 35. *Claude Code is open and signed in: version, plan and folder at the top, the prompt at the bottom.*

> **Want to see the sign-in again?** Inside Claude Code type `/logout`, then run `claude` again. It asks you to sign in as it did on the first run.